

⑤ Square roots and Cube roots

square root :- If $x^2 = y$, we say that the square root of y is x and we write,

$$\sqrt{y} = x.$$

$$\text{Thus } \sqrt{4} = 2$$

$$\sqrt{9} = 3$$

$$\sqrt{196} = 14.$$

cube root :- The cube root of a given number x is the number whose cube is x . We denote the cube root of x

by $\sqrt[3]{x}$.

$$\text{Thus } \sqrt[3]{8} = \sqrt[3]{2 \times 2 \times 2} = 2$$

$$\sqrt[3]{343} = \sqrt[3]{7 \times 7 \times 7} = 7$$

$$\star \sqrt{xy} = \sqrt{x} \cdot \sqrt{y}$$

$$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}} = \frac{\sqrt{x}}{\sqrt{y}} \cdot \frac{\sqrt{y}}{\sqrt{y}} = \frac{\sqrt{xy}}{y}$$

ex. $\sqrt{6084}$ Evaluate

$$\begin{aligned} 6084 &= 2 \times 2 \times 3 \times 3 \times 13 \times 13 \\ &= 2^2 \times 3^2 \times 13^2 \end{aligned}$$

$$\begin{aligned} \sqrt{6084} &= 2 \times 3 \times 13 \\ &= 2 \times 39 \\ &= 78 \end{aligned}$$

Ans.

2	6084
2	3042
3	1521
3	507
13	169
13	13
	1

Ex. ②

Find square root of

$$\sqrt{1471369}$$

Solⁿ:-

		1 2 1 3	
	1	147	1369
+ 1	1	-1	
	22	47	
+ 2	2	-44	
	241	313	
+ 1	1	-241	
	2423	7269	
		-7269	
		X	

$$\sqrt{1471369} = 1213 \quad \underline{\text{Ans.}}$$

Ex ③ Evaluate

$$\sqrt{248 + \sqrt{51 + \sqrt{169}}}$$

Solⁿ:- $\sqrt{248 + \sqrt{51 + 13}}$

$$= \sqrt{248 + \sqrt{64}}$$

$$= \sqrt{248 + 8}$$

$$= \sqrt{256}$$

$$= 16$$

Ans.

Ex. ⑥ what is the square root of 0.0009 ?

Solⁿ:-
$$\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{\sqrt{9}}{\sqrt{10000}}$$
$$= \frac{3}{100} = 0.03 \quad \underline{\text{Ans.}}$$

Ex. ⑧ $\sqrt{86.49} + \sqrt{5 + (?)^2} = 12.3$ (RR B 2002)

Solⁿ:- $\sqrt{86.49} + \sqrt{5 + x^2} = 12.3$

$$\Rightarrow 9.3 + \sqrt{5 + x^2} = 12.3$$

$$\Rightarrow \sqrt{5 + x^2} = 12.3 - 9.3$$

$$\Rightarrow \sqrt{5 + x^2} = 3$$

$$\Rightarrow 5 + x^2 = 9$$

$$\Rightarrow x^2 = 4$$

$$\Rightarrow x = \pm 2 \quad \underline{\text{Ans.}}$$

Ex. ⑨ find $\sqrt{\frac{0.289}{0.00121}}$ (IGNOU 2003)

Solⁿ:-
$$\sqrt{\frac{0.289}{0.00121}} = \sqrt{\frac{0.28900}{0.00121}} = \sqrt{\frac{28900}{121}}$$
$$= \frac{170}{11} \quad \underline{\text{Ans.}}$$

Ex. ⑩ If $\sqrt{1 + \frac{x}{144}} = \frac{13}{12}$, find x

Solⁿ:- $1 + \frac{x}{144} = \frac{169}{144}$

$$\Rightarrow \frac{x}{144} = \frac{25}{144} \Rightarrow x = 25 \quad \underline{\text{Ans.}}$$

How to find square root values:

$$\sqrt{2} = 1.41$$

$$\sqrt{3} = 1.73$$

$$\sqrt{4} = 2 \quad \text{perfect square}$$

$$\sqrt{5} = 2.23$$

$$\sqrt{6} = 2.44 \quad \text{imperfect square}$$

$$\sqrt{7} = 2.64$$

*
 $\sqrt{64}$

2	64
2	32
2	16
2	8
2	4
2	2
	1

$$64 = \underbrace{2 \times 2} \times \underbrace{2 \times 2} \times \underbrace{2 \times 2}$$

$$\sqrt{64} = 2 \times 2 \times 2 = 8 \quad \text{Ans.}$$

Long Division method:-

		<u>25</u> → <u>Ans</u>
2		<u>625</u>
+ 2	- 4	
45		225
	- 225	
		0

$$\sqrt{625} = 25$$

Ans.

41
x 1
41
42
x 2
84
45
x 5
225

$$\sqrt{5} = 2.23$$

2.23

Ans.

2	5
+ 2	- 4
42	100
+ 2	- 84
443	1600
+ 3	- 1329
446	271

$$\begin{array}{r} 41 \\ \times 1 \\ \hline 41 \end{array}$$

$$\begin{array}{r} 42 \\ \times 2 \\ \hline 84 \end{array}$$

$$\begin{array}{r} 441 \\ \times 1 \\ \hline 441 \end{array}$$

$$\begin{array}{r} 442 \\ \times 2 \\ \hline 884 \end{array}$$

$$\begin{array}{r} 443 \\ \times 3 \\ \hline 1329 \end{array}$$

$$\begin{array}{r} 444 \\ \times 4 \\ \hline 1776 \end{array}$$

104

$$\sqrt{18}$$

$$= \sqrt{3 \times 3 \times 2}$$

$$= 3\sqrt{2}$$

$$= 3 \times 1.41$$

$$= 4.23$$

Ans.

2	18
- 3	9
3	3
	1

$$\sqrt{2848}$$

	53.36
5	2848
+ 5	- 25
103	348
+ 3	- 309
1063	3900
+ 3	- 3189
10666	71100
+ 6	- 63996
10672	

$$\begin{array}{r} 103 \\ \times 3 \\ \hline 309 \end{array}$$

$$\begin{array}{r} 1063 \\ \times 3 \\ \hline 3189 \end{array}$$

$$\begin{array}{r} 10662 \\ \times 2 \\ \hline 21324 \end{array}$$

$$\begin{array}{r} 10666 \\ \times 6 \\ \hline 63996 \end{array}$$